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Mendelian Genetics in Man

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Mendelian Inheritance in Man

- Mendelian inheritance is a characteristic for most of the traits
  - which are monogenic or single factor traits
  - when environmental conditions do not influence the phenotypic expression of the genes.
- However, there are a number of exceptions to the rules of Mendelian inheritance in human genetics

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Non-Mendelian Inheritance

Inheritance	Example
Polygenic or Polygenetic Inheritance	Autism, Hypertension, Stature
Incomplete Dominance aka Intermediate Expression	Tay Sachs, Voice Pitch
Codominance	Sickle Cell Trait, AB Blood Type
Multiple Allele Series	ABO Blood Group, HLAs
Modifying or Regulator Genes	Cataract, Homeotic Genes
Incomplete Penetrance	Multiple Sclerosis
Pleiotropy	Albinism
Epistatis	Dermatoglyphics
Sex Related Genetic Effect	Hemophilia
Environmental Influences	Intelligence

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Polygenic Traits

- Polygenic or polygenetic traits are influenced by more than one pair of genes
- Each of the alleles will have an additive effect (+ or -)
- These are also known as **continuous traits or quantitative traits**
- **Ex. Human Stature** – the combined size of all the body parts from head to foot determine the height of an individual and the parts' individual sizes are in turn influenced by many genes and also environment (Growth Hormone, Nutrition, Emotions etc.)
- Human diseases like autism, hypertension, glaucoma

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Intermediate Expression / Incomplete Dominance

- There is an apparent blending in the phenotype in heterozygous individuals - Ex. Red X White = Pink
- In humans, the male voice pitch is an intermediate expression – the lowest pitch is AA and highest pitch is aa while the intermediate baritones are heterozygous Aa
- The child killer disease Tay Sachs (fluid pressure on brain and subsequent degeneration) is characterized by incomplete dominance where the heterozygous individuals are genetically programmed to produced only 40-60% of the enzyme hexosaminidase A.

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Codominance

- Both alleles express themselves in heterozygous condition and the phenotype is not an intermediate - Ex. AB blood type and Sickle Cell Trait common in African Americans
- The gene for Hemoglobin Hb has two alleles – HbA (normal) and HbS (mutated gene) and three phenotypes
  - **HbAHbA** – Normal Phenotype and all RBCs and hemoglobin are normal
  - **HbAHbS** – Sickle Cell Trait – 50% of hemoglobin in every RBC is abnormal and RBCs are slightly distorted / suffer only under hypoxia / resistant to Malaria
  - **HbSHbS** – Sickle Cell Anemia - 100% abnormality in hemoglobin and all RBCs in sickle shape - fatal

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Sickle Cell RBCs



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Multiple-Allele Series

- A trait can be controlled by more than just a **pair** of alleles
- Ex. ABO system – there are three alleles A, B and O but the individual is inheriting only two of them!
- Some traits are believed to be controlled by far more than two alleles – Ex. HLA system (Human Leukocyte Antigen)
- HLA - identifying and rejecting foreign tissue by regulating antibody production
- Known to have 30,000,000 different genotypes
- Play a very important role in rejection of organ transplant

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Modifying & Regulator Genes

- **Modifying Genes** alter how certain other genes are expressed in a phenotype – Ex. Cataract Gene is dominant but produces varying degree of vision impairment depending on the presence of specific allele on **companion modifying gene** (also environment like alcoholism, diabetes etc)
- **Regulator Genes** – initiate or block expression of phenotype. Ex. Time of production of specific proteins or new structural parts during aging – Homeotic Genes

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Incomplete Penetrance

- Incomplete penetrance is a situation where the effect of a particular gene does not normally occur unless there are certain environmental factors present
- Ex. One may inherit the gene for Type 2 Diabetes but can avoid it unless they are obese, stressful and insomniac
- **Multiple Sclerosis** (a progressive neuro-degenerative disorder leading to paralysis) – Triggered due to a specific virus called Epstein Barr Virus

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Pleiotropy

- A single gene may be responsible for a variety of traits
- Pleio – More; Tropy – Convert
- **Albinism** – The gene for this trait not only results in typical melanin deficiency of skin, hair and eyes, but also cause defects in vision
- **Sickle Cell Trait** is another example – can also cause hearth, lung, kidney and vision problems, strokes, osteoporosis etc.

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Epistatis

- Interaction between **genotypes at two different genetic loci** Epi – Over; Statis – to sit – "To Sit on Top"
- Exception to Mendel's **Independent Assortment – 9:3:3:1 integrity is lost**
- Ex. If gene at one locus determines if the color should be red or white, another gene at a different locus determines if the pigment should be produced or not!
- Dermatoglyphics in man are believed to be epistatic

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Sex Related Genetic Effects

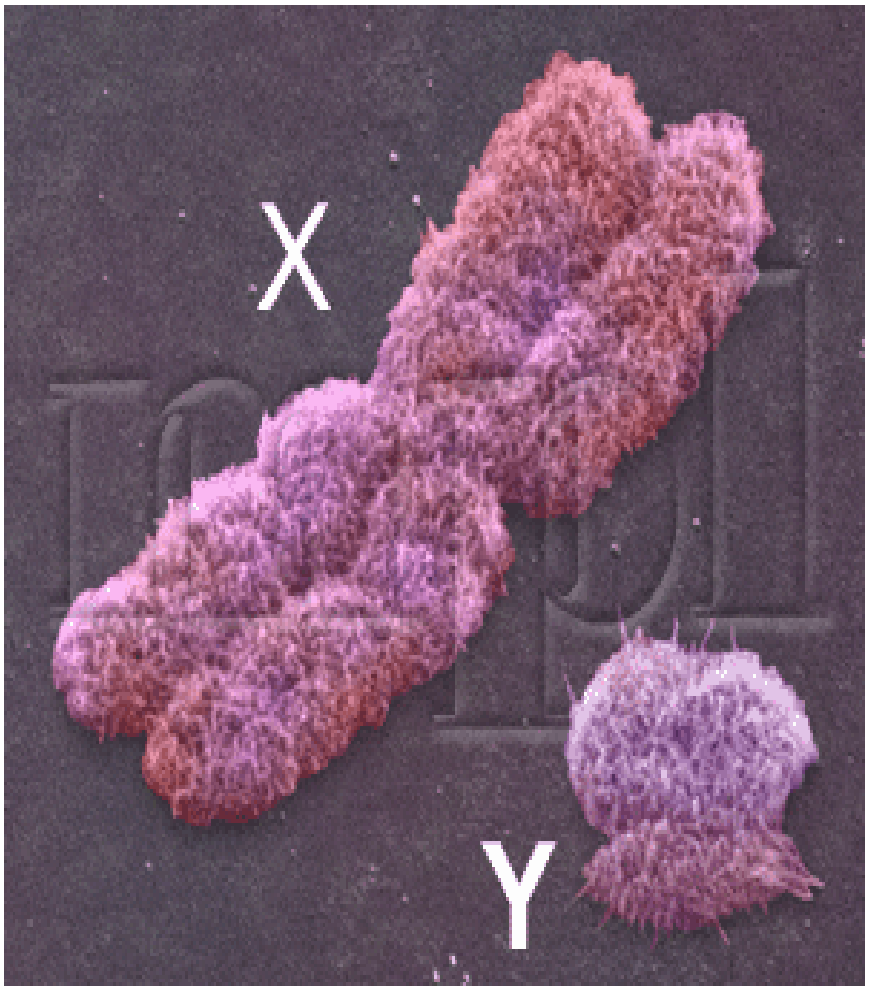
- Three types
- **Sex Limited Genes:** Inherited by both men and women but are normally expressed in the phenotype of one of them – Ex. Heavy male beard (gene for facial hair)
- **Sex Controlled Genes:** Expressed in both sexes differently – Ex. Gout – Joint swellings – Men 8 times more likely to have severe symptoms
- **Genome Imprinting:** Genes express differently depending on the sex of the parent from whom they are inherited

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Sex-linked Inheritance

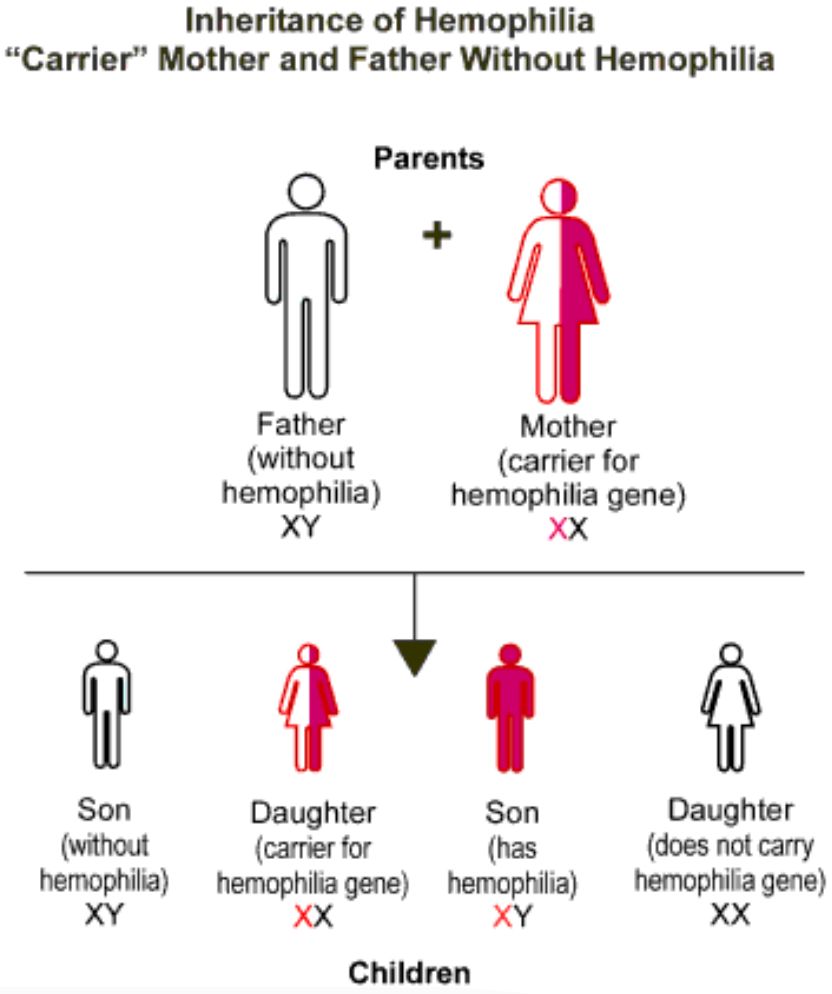
- Besides sex information, there are 80,000 genes in X and 90 in the Y chromosome
- Traits linked to X and Y are referred to as "X-linked" and "Y-linked" respectively
- Most of the X-linked genes are recessive – Colorblindness, Hemophilia, Baldness



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Sex-linked Inheritance



- Men more likely to show them because they are **hemizygous** – when only one copy of the gene is present
- Women who are homozygous also show the traits

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Environmental Influences

- Environmental factors alter phenotypes
- Abnormal growth and conditions affecting growth
  - Nutrition
  - Oxygen deprivation during pregnancy, childbirth – Autism?
  - Habits – smoking can cause some mutations
  - Radiation exposures etc.

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Lethal Gene Action

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Lethal Gene Action

- One of the most important assumptions in genetics is equal survival of all the gametes and zygotes that are produced as a result of segregation
- This assumption is true in majority of cases
- Some genes affect the survival of the gamete or the zygote (the individual) if present in specific genotypes
- Their effect ranges from improving survival to no effect to even death

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Lethal Gene Action

- Based on their effect on survival, genes can be categorized into
  - Lethal
  - Semi Lethal
  - Sub Vital
  - Vital
  - Super Vital

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Lethal Gene

- A lethal gene **causes death of all individuals** carrying it in the appropriate genotype before they reach adulthood
- The appropriate genotype for an allele would depend on its dominance relationship with other alleles
- The stage of development at which a lethal gene produces its lethal affect varies considerably – some cause death very early in development while others allow survival of individuals to reproductive age

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Lethal Genes

- Lethal genes can be further classified into **Recessive Lethals, Dominant Lethals, Conditional Lethals and Gametic Lethals**
- **Recessive Lethals:**
  - A majority of genes are recessive lethals
  - Lethality is expressed only in homozygous condition
  - There are some genes which are recessive, but still show dominant phenotypic effect even in heterozygous condition – Xeroderma Pigmentosum

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Lethal Genes

- **Dominant Lethals**
  - These lethal genes reduce viability in the heterozygous conditions as well
  - Ex. Epiloia in man – This gene causes abnormal skin growths, severe mental defects and multiple tumors in the heterozygous condition, so that they die before reaching adulthood
  - Dominant lethals cannot be maintained in the population while the recessive lethals are maintained in the heterozygous state
  - Dominant lethals have to be produced in every generation through mutation

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Lethal Genes

- **Conditional Lethals**
  - Some lethal genes allow normal development and survival, except under a specified condition which is necessary for their lethal effect
  - Such lethal genes that require a specific condition for their lethal action are termed conditional lethals
  - Ex. Diabetes, Xeroderma Pigmentosum (light, nutrition)

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Gametic Lethals

- These lead to inviability of gametes or make them incapable of fertilization
- Gametic lethals lead to a drastic departure from the typical ratio expected in a segregating generation – this phenomenon is called '*SEGREGATION DISTORTION* OR *MEIOTIC DRIVE*'
- They are different from Zygotic Lethals

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Other Lethal Gene Types

- **Semi Lethal –** Cause the death of more than 90% of the individuals. Only less than 10% of the individuals survive
- **Sub Vital Genes –** Kill less than 90% of the individuals who carry them

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Vital Genes

- These genes **do not affect the survival** of the individuals in whom they are present – they neither enhance nor reduce the viability of the individuals carrying them
- It does not imply that these genes are no necessary for the survival of the concerned organism
- It simply means that the survival of the organism is not affected by the fact that the concerned vital genes are present in homozygous or heterozygous state

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Super Vital Genes

- These mutant genes **enhance the survival** of those individuals that carry them in the appropriate genotype as compared to that of the wild type (naturally occurring) allele
- **Ex.** Disease resistant crops – high productivity etc.

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Multifactor Hypothesis & Quantitative Genetics

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Quantitative Genetics

- Genetics is the study of variation and without variation genetic analysis at any level becomes difficult
- In many cases, this genetic analysis of **variation involves striking differences** between the organisms – Red or White, Tall or Short
- In other cases, however, biological variation is difficult to categorize like height and weight in humans are traits that vary continuously in the population
- Distinct cases cannot be identified for such traits, making it difficult to analyze these cases by conventional genetic methods
- Quantitative genetics is the discipline that deals with these complex traits

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Quantitative Traits

- Quantitative genetics is based on measurements of individuals within a population of organisms
- It recognize two important factors
  - **Most quantitative traits involve contribution of many different genes**. A single gene may exert primary influence but usually there are many equally influential genes
  - **Quantitative traits are often influenced by environmental factors**. A quantitative trait will show phenotypic variation over and above that caused by genotypic differences

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Multifactor Hypothesis

- It was difficult to reconcile Mendelian Genetics with quantitative traits
- **R.A. Fisher – Multifactor Hypothesis -** ***"Quantitative traits are not determined by single gene but many and their alleles each has small and approximately additive effects"***
- Thus, a phenotype of an individual depended on its genotype at all the relevant loci, with each allele adding or subtracting a small amount

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Polygenic Traits

- These traits are **Polygenic Traits**
- Fisher also proposed that many environmental factors influenced the trait by adding or subtracting effects in a manner similar to that of all the genetic loci
- This combination of multiple environmental and genetic factors determined the phenotype of an individual

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Significance of Multifactor Hypothesis

1. It readily explains variability among phenotypes of quantitative traits through large number of genotypes and environmental factors
2. Explains why offspring of parents who are phenotypically distinct often lead to intermediate types
3. Explains why intermediate phenotypes are more common than extreme phenotypes which are relatively rare, because of the operation of multiple genotypes having plus or minus effect.

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